Page 1

LIZA 0.03mg/3mg
film-coated tablets

COMPOSITION

Each film-coated tablet contains 0.03 mg of ethinylestradiol and 3 mg of drospirenone.

PHARMACEUTICAL FORM

Round, yellow, film-coated tablets.

CLINICAL PARTICULARS

Therapeutic indications
Oral contraception.

POSOLOGY AND METHOD OF ADMINISTRATION

For oral administration

How to take LIZA film tablets?

The tablets must be taken every day at about the same time, if necessary with a little liquid in the order shown on blister pack. One tablet is to be taken daily for 21 consecutive days. Each subsequent pack is started after 7-day tablet-free interval, during which time withdrawal bleed occurs. This usually starts on day 2-3 after the last tablet and may not have finished before the next pack is started.

How to start LIZA film tablets?

- No preceding hormonal contraceptive use (in the past month)

Tablet-taking has to start on day 1 of the woman's natural cycle (i.e.the first day of her menstrual bleeding).

- Changing from a combined oral contraceptive (COC), vaginal ring, or transdermal patch

The woman should start with LIZA preferably on the day after the last active tablet (the last tablet containing the active substances) of her previous combined COC, but at the latest on the day following the usual tablet-free interval of her previous COC. In case a vaginal ring or transdermal patch has been used, the woman should start using LIZA preferably on the day of removal, but at the latest when the next application would have been due.

- Changing from a progestogen-only-method (progestogen-only pill, injection, implant) or from a progestogen-releasing intrauterine system (IUS)

The woman may switch any day from the progestogen-only pill (from an implant or the IUS on the day of its removal, from an injectable when the next injection would be due) but should in all of these cases be advised to additionally use a barrier method for the first 7 days of tablet-taking.

- Following first trimester abortion

The woman may start immediately. When doing so, she need not take additional contraceptive measures.

- Following delivery or second-trimester abortion

Woman should be advised to start at day 21 to 28 after delivery or second-trimester abortion. When starting later, the woman should be advised to additionally use a barrier method for the first 7 days. However, if intercourse has already occurred, pregnancy should be excluded before the actual start of COC use or the woman has to wait for her first menstrual period.

For breast feeding women

See Section Pregnancy & lactation.

Management of missed tablets

The following advice only refers to missed active tablets (rows 1- 3 of the blister):

If the user is **less than 12 hours** late in taking any tablet, contraceptive protection is not reduced. The woman should take the tablet as soon as she remembers and should take further tablets at usual time.

If she is **more than 12 hours** late in taking any tablet, contraceptive protection may be reduced. The management of missed tablets can be guided by the following two basic rules:

1. tablet-taking must never be discontinued for longer than 7 consecutive days.
2. 7 days of uninterrupted tablet-taking are required to attain adequate suppression of the hypothalamic-pituitary- ovarian-axis.

Accordingly the following advice can be given in daily practice:

- **Week 1**

The user should take the last missed tablet as soon as she remembers, even if this means taking two tablets at the same time. She then continues to take the remaining tablets at her usual time. In addition, a barrier method such as condom should be used for the next 7days. If intercourse took place in the preceding 7 days, the possibility of pregnancy should be considered. The more tablets are missed and the closer they are to the regular tablet-free interval, the higher the risk of a pregnancy.

- **Week 2**

The user should take the last missed tablet as soon as she remembers, even if this means taking two tablets at the same time. She then continues to take the remaining tablets at her usual time. Provided that the woman has taken her tablets correctly in the 7 days preceding the first missed tablet, there is no need to use extra contraceptive precautions. However, if she has missed more than1 tablet, the woman should be advised to use extra precautions for the next 7 days following the last forgotten tablet.

- **Week 3**

The risk of reduced reliability is imminent because of the forthcoming 7 day tablet-free interval.

However, by adjusting the tablet-intake schedule, reduced contraceptive protection can still be prevented. By adhering to either of the following two options, there is therefore no need to use extra contraceptive precaution, provided that in the 7 days preceding the first missed tablet the woman has taken all tablets correctly. If this is not the case, she should follow the first of these two options and use extra precautions for the next 7 days as well.

1. The user should take the last missed tablet as soon as she remembers, even if this means taking two tablets at same time. She then continues to take remaining tablets at her usual time. The next blister pack must be started as soon as the current blister pack is finished i.e.no gap should be left between packs. The user is unlikely to have a withdrawal bleed until the end of the second pack, but she may experience spotting or breakthrough bleeding on tablet-taking days.
2. The woman may also be advised to discontinue tablet-taking from the current blister pack. She should then have a tablet-free interval of up to 7 days, including the days she missed tablets, and subsequently continue with the next blister pack.

If the woman missed tablets and subsequently has no withdrawal bleed in the first normal tablet–free interval, the possibility of a pregnancy should be considered.

Advise in case of gastro-intestinal disturbances

In case of severe gastro-intestinal disturbances (e.g.,vomiting or severe diarrhoea), absorption may not be complete and additional contraceptive measures should be taken. If vomiting occurs within 3-4 hours after tablet taking, a new (replacement) tablet should be taken as soon as possible. The new tablet should be taken within 12 hours of the usual time of tablet-taking if possible.
If more than 12 hours elapse, the advice concerning missed tablets, as give in section "Management of missed tablets" is applicable. If the woman does not want to change her normal tablet-taking schedule, she has to take the extra tablet(s) from another blister pack.

How to postpone a withdrawal bleed

To delay a period the woman should continue with another blister pack of LIZA without a tablet-free interval. The extension can be carried on for as long as wished until the end of the second pack. During the extension the woman may experience breakthrough-bleeding or spotting. Regular intake is then resumed after the usual 7-days tablet –free interval.

To shift her periods to another day of the week than the woman is used to with her current scheme, she can be advised to shorten her forthcoming tablet-free interval by as many days as she likes. The shorter the interval, the higher the risk that she does not have a withdrawal bleed and will experience breakthrough-bleeding and spotting during subsequent pack (just as when delaying period).

CONTRAINDICATIONS

Combined oral contraceptives (COC) should not be used in the presence of any the conditions listed below. Should any of the conditions appear for the first time during COC use, the product should be stopped immediately.

- Venous thrombosis present or in history (deep venous thrombosis, pulmonary embolism).
- Arterial thrombosis present or in history (e.g. myocardial infarction) or prodromal conditions (e.g. angina pectoris and transient ischaemic attack).
- Cerebro-vascular accident present or in history.
- The presence of a severe or multiple risk factor(s) for arterial thrombosis.
  - Diabetes mellitus with a vascular symptoms
  - Severe hypertension
  - Severe dyslipoproteinemia
- Hereditary or acquired predisposition for venous or arterial thrombosis, such as APC-resistance, antithrombin-III-deficiency, protein C deficiency, protein S deficiency, hyperhomocysteinemia and antiphospholipid-antibodies (anticardiolipin-antibodies, lupus anticoagulant).
- Pancreatitis or a history thereof if associated with severe hypertriglyceridemia.
- Presence or history of severe hepatic disease as long as liver function values have not returned to normal.
- Severe renal insufficiency or acute renal failure.
- Presence or history of liver tumours (benign or malignant).
- Known or suspected sex-steroid influenced malignancies (e.g. of the genital organs or the breasts).
- History of migraine with focal neurological symptoms.
- Hypersensitivity to the active substances or to any of the excipients of LIZA film- coated tablets.

LIZA is contraindicated for concomitant use with the medicinal products containing ombitasvir / paritaprevir / ritonavir and dasabuvir (See Section Warnings and Precautions and Section Interactions with Other Medicaments).

SPECIAL WARNINGS AND PRECAUTIONS FOR USE

Warnings

If any of the conditions/risk factors mentioned below are present, the benefits of COC use should be weighed against the possible risks for each individual woman and discussed with the woman before she decides to start using it. In the event of aggravation, exacerbation or first appearance of any of these conditions or risk factors, the woman should contact her physician. The physician should then decide on whether COC use should be discontinued.

- **Circulatory Disorders**

The use of any combined oral contraceptive carries an increased risk of venous thromboembolism (VTE) compared with no use. The excess risk of VTE is highest during the first year a woman ever uses a combined oral contraceptive. Epidemiological studies have also associated the use of combined COCs with an increased risk for arterial (myocardial infarction, transient ischaemic attack) thromboembolism. Extremely rarely, thrombosis has been reported to occur in other blood vessels, e.g. hepatic, mesenteric, renal, cerebral or retinal veins and arteries, in contraceptive pill users. There is no consensus as to whether the occurrence of these events is associated with the use of hormonal contraceptives.

Symptoms of venous or arterial thrombotic/thromboembolic events or of a cerebrovascular accident can include:
- unusual unilateral leg pain and/ or swelling
- sudden severe pain in the chest, whether or not it radiates to the left arm
- sudden breathlessness
- sudden onset of coughing without a clear cause
- any unusual, severe, prolonged headache
- sudden partial or complete loss of vision
- diplopia
- slurred speech or aphasia
- vertigo
- collapse with or without focal seizure
- weakness or very marked numbness suddenly affecting one side or one part of the body
- motor disturbances
- 'acute abdomen'.

The risk for venous thromboembolic complications in COC users increases with:
- increasing age.
- a positive family history (venous thromboembolism ever in a sibling or parent at relatively early age). If a hereditary predisposition is suspected, the woman should be referred to a specialist for advice before deciding about any COC use.
- prolonged immobilisation, major surgery, any surgery to the legs, or major trauma. In these situations it is advisable to discontinue the pill (in the case of elective surgery at least four weeks in advance) and not resume until two weeks after complete remobilisation. Antithrombotic treatment should be considered if the pills have not been discontinued in advance.
- obesity (body mass index over 30 kg/m²).
- there is no consensus about the possible role of varicose veins and superficial thrombophlebitis in the onset or progression of venous thrombosis.

The risk of arterial thromboembolic complications or of a cerebrovascular accident in COC users increases with:
- increasing age.
- smoking (women over 35 years should be strongly advised not to smoke if they wish to use an COC).
- dyslipoproteinemia.
- hypertension.
- migraine.
- obesity (body mass index over 30 kg/m²).
- valvular heart disease.
- atrial fibrillation.

The presence of one serious risk factor or multiple risk factors for venous or arterial disease, respectively, can also constitute a contra-indication. The possibility of anticoagulant therapy should also be taken into account. COC users should be specifically pointed out to contact their physician in case of possible symptoms of thrombosis. In case of suspected or confirmed thrombosis, COC use should be discontinued. Adequate alternative contraception should be initiated because of the teratogenicity of anticoagulant therapy (coumarins).

The increased risk of thromboembolism in the puerperium must be considered (for information on Pregnancy & Lactation). Other medical conditions which have been associated with adverse vascular events include diabetes mellitus, systemic lupus erythematosus, haemolytic uremic syndrome and chronic inflammatory bowel disease (Crohn's disease or ulcerative colitis) and sickle cell disease.

An increase in frequency or severity of migraine during COC use (which may be prodromal of a cerebrovascular event) may be a reason for immediate discontinuation of the COC.

- **Tumours**

An increased risk of cervical cancer in long-term users of COCs has been reported in some epidemiological studies, but there continues to be controversy about the extent to which this finding is attributable to the confounding effects of sexual behaviour and other factors such as human papilloma virus (HPV).

In rare cases, benign liver tumours, and even more rarely, malignant liver tumours have been reported in users of COCs. In isolated cases, these tumours have led to life-threatening intra-abdominal haemorrhages. A hepatic tumour should be considered in the differential diagnosis when severe upper abdominal pain, liver enlargement or signs of intra-abdominal haemorrhage occur in women taking COCs.

- **Other conditions**

The progestin component in LIZA is an aldosterone antagonist with potassium sparing properties. In most cases, no increase of potassium levels is to be expected. In a clinical study, however in some patients with mild or moderate renal impairment and concomitant use of potassium-sparing medicinal products serum potassium levels slightly, but not significantly, increased during drospirenone intake. Therefore, it is recommended to check serum potassium during the first treatment cycle in patients presenting with renal insufficiency and a pretreatment serum potassium in the upper reference range, and particularly during concomitant use of potassium sparing medicinal products. See also section Drug Interactions.

Women with hypertriglyceridemia, or a family history thereof, may be at an increased risk of pancreatitis when using COCs. Although small increases in blood pressure have been reported in many women taking COCs, clinically relevant increases are rare. Only in these rare cases an immediate discontinuation of COC use is justified. If, during the use of a COC in pre-existing hypertension, constantly elevated blood pressure values or a significant increase in blood pressure do not respond adequately to antihypertensive treatment, the COC must be withdrawn. Where considered appropriate, COC use may be resumed if normotensive values can be achieved with antihypertensive therapy.

The following conditions have been reported to occur or deteriorate with both pregnancy and COC use, but the evidence of an association with COC use is inconclusive: jaundice and/or pruritus related to cholestasis; gallstones; porphyria; systemic lupus erythematosus; haemolytic uremic syndrome; Sydenham's chorea; herpes gestationis; otosclerosis-related hearing loss.

In women with hereditary angioedema exogenous estrogens may induce or exacerbate symptoms of angioedema.

Acute or chronic disturbances of liver function may necessitate the discontinuation of COC use until markers of liver function return to normal. Recurrence of cholestatic jaundice and/or cholestasis-related pruritus which previously occurred during pregnancy or during previous use of sex steroids necessitates the discontinuation of COCs.

Although COCs may have an effect on peripheral insulin resistance and glucose tolerance, there is no evidence for a need to alter the therapeutic regimen in diabetics using low-dose COCs (containing < 0.05 mg ethinylestradiol). However, diabetic women should be carefully observed, particularly in the early stage of COC use.

Worsening of endogenous depression, of epilepsy, of Crohn's disease and of ulcerative colitis has been reported during COC use.

Chloasma may occasionally occur, especially in women with a history of chloasma gravidarum. Women with a tendency to chloasma should avoid exposure to the sun or ultraviolet radiation whilst taking COCs.

Medical examination/consultation

Prior to the initiation or reinstitution of LIZA, a complete medical history (including family history) should be taken and pregnancy must be ruled out. Blood pressure should be measured and a physical examination should be performed, guided by the contraindications and warnings. The woman should also be instructed to carefully read the user leaflet and to adhere to the advice given. The frequency and nature of examinations should be based on established practice guidelines and be adapted to the individual woman.

Women should be advised that oral contraceptives do not protect against HIV infections (AIDS) and other sexually transmitted diseases.

Reduced efficacy

The efficacy of COCs may be reduced in the event of e.g. missed tablets (see **Posology and method of administration**), gastro-intestinal disturbances vomiting or severe diarrhea (see **Posology and method of administration**) or concomitant medication (see **Interactions with other medicinal products and other forms of interaction**).

Reduced cycle control

During the use of any COCs, irregular bleeding (spotting or breakthrough bleeding) may occur, especially during the first months of use. Therefore, the evaluation of any irregular bleeding is only meaningful after an adaptation interval of about three cycles.

If bleeding irregularities persist or occur after previously regular cycles, then non-hormonal causes should be considered and adequate diagnostic measures are indicated to exclude malignancy or pregnancy. These may include curettage.

In some women withdrawal bleeding may not occur during the tablet-free interval. If the COC has been taken according to the directions described in section **Posology and method of administration**, it is unlikely that the woman is pregnant. However, if the COC has not been taken according to these directions prior to the first missed withdrawal bleed or if two withdrawals bleeds are missed, pregnancy must be ruled out before COC use is continued.

LIZA 0.03mg/3mg film-coated tablets : The yellow tablets of this medicinal product contains 62mg of lactose per tablet.

Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption who are on a lactose-free diet should take this amount into consideration.

ALT elevations

During clinical trials with patients treated for hepatitis C virus infections (HCV) with the medicinal products containing ombitasvir / paritaprevir / ritonavir and dasabuvir with / without ribavirin, transaminase (ALT) elevations higher than 5 times the upper limit of normal (ULN) occurred significantly more frequent in women using ethinylestradiol-containing medications such as combined hormonal contraceptives (CHCs). Patients who are taking ethinylestradiol-containing medicinal products must switch to an alternative method of contraception (e.g. progestin only contraception or non-hormonal methods) prior to initiating ombitasvir / paritaprevir / ritonavir and dasabuvir therapy (See Section Contraindications and Section Interactions with Other Medicaments).

INTERACTIONS WITH OTHER MEDICINAL PRODUCTS AND OTHERS FORMS OF INTERACTION

Note: The prescribing information of concomitant medications should be consulted to identify potential interactions.

Influence of other medicinal products on LIZA

Interactions between oral contraceptives and other medicinal products may lead to breakthrough bleeding and/or contraceptive failure. The following interactions have been reported in the literature.

This has been established with hydantoins, barbiturates, primidone, carbamazepine and rifampicin; oxcarbazepine, topiramate, felbamate, ritonavir, griseofulvin and the herbal remedy St. John's Wort (hypericum perforatum) are also suspected. The mechanism of this interaction appears to be based on the hepatic enzyme-inducing properties of these active substances. Maximal enzyme induction is generally not seen for 2-3 weeks but may then be sustained for at least 4 weeks after the cessation of drug therapy.

Contraceptive failures have also been reported with antibiotics, such as ampicillin and tetracyclines. The mechanism of this effect has not been elucidated.

Women on short-term treatment (up to one week) with any of the above-mentioned classes of medicinal products or individual active substances should temporarily use a barrier method in addition to the COC, i.e. during the time of concomitant medicinal product administration and for 7 days after their discontinuation.

For women on rifampicin a barrier method should be used in addition to the COC during the time of rifampicin administration and for 28 days after its discontinuation.

If concomitant medicinal product administration runs beyond the end of the tablets in the COC blister pack, the next COC pack should be started without the usual tablet-free interval. The placebo tablets must be discarded and the next COC pack should be started right away.

In women on long-term treatment with hepatic enzyme-inducing drugs, experts have recommended to increase the contraceptive steroid doses.

If a high contraceptive dosage is not desirable or appears to be unsatisfactory or unreliable, e.g. in the case of breakthrough bleeding, another non-hormonal contraception method should be advised.

The main metabolites of drospirenone in human plasma are generated without involvement of the cytochrome P450 system. Inhibitors of this enzyme system are therefore unlikely to influence the metabolism of drospirenone.

Influence of LIZA 0.03mg/3mg film-coated tablet on other medicinal products

Oral contraceptives may affect the metabolism of certain other active substances. Accordingly, plasma and tissue concentrations may either increase (e.g. cyclosporin) or decrease (e.g. lamotrigine).
Based on in vitro inhibition studies and in vivo interaction studies in female volunteers using omeprazole, simvastatin and midazolam as marker substrate, an interaction of drospirenone at doses of 3 mg with the metabolism of other active substances is unlikely.

Other interactions

In patients without renal insufficiency, the concomitant use of drospirenone and ACE-inhibitors or NSAIDs did not show a significant effect on serum potassium. Nevertheless, concomitant use of LIZA 0.03mg/3mg film-coated tablet with aldosterone antagonists or potassium-sparing diuretics has not been studied. In this case, serum potassium should be tested during the first treatment cycle. See also section **Warnings**.

Laboratory tests

The use of contraceptive steroids may influence the results of certain laboratory tests, including biochemical parameters of liver, thyroid, adrenal and renal function, plasma levels of (carrier) proteins, e.g. corticosteroid-binding globulin and lipid/lipoprotein

Page 2

fractions, parameters of carbohydrate metabolism and parameters of coagulation and fibrinolysis. Changes generally remain within the normal laboratory range. Drospirenone causes an increase in plasma renin activity and plasma aldosterone induced by its mild antimineralocorticoid activity.

Concomitant use with the medicinal products containing ombitasvir / paritaprevir / ritonavir and dasabuvir, with or without ribavirin may increase the risk of ALT elevations (See Section Contraindications and Section Warnings and Precautions). Therefore, users must switch to an alternative method of contraception (e.g., progestogen-only contraception or non-hormonal methods) prior to starting therapy with this combination drug regimen. LIZA can be restarted 2 weeks following completion of treatment with this combination drug regimen.

PREGNANCY AND LACTATION

The product is not indicated during pregnancy.

If pregnancy occurs during use of the product, the preparation should be withdrawn immediately. Extensive epidemiological studies have revealed neither an increased risk of birth defects in children born to women who used COCs prior to pregnancy, nor a teratogenic effect when COCs were taken inadvertently during pregnancy. No such studies are conducted with the product. Animal studies have shown undesirable effects during pregnancy and lactation. Based on these animal data, undesirable effects due to hormonal action of the active compounds cannot be excluded. However, general experience with COCs during pregnancy did not provide evidence for an actual undesirable effect in humans.

The available data regarding the use of the product during pregnancy are too limited to permit conclusions concerning negative effects of the product on pregnancy, health of the foetus or neonate. To date, no relevant epidemiological data are available.

Lactation may be influenced by COCs as they may reduce the quantity and change the composition of breast milk. Therefore, the use of COCs should generally not be recommended until the breast-feeding mother has completely weaned her child. Small amounts of the contraceptive steroids and/or their metabolites may be excreted with the milk during COC use. These amounts may affect the child.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

No studies on the effects on the ability to drive and use machines have been performed. No effects on ability to drive and use machines have been observed in users of COCs.

UNDESIRABLE EFFECTS

The following adverse drug reactions have been reported during use of LIZA:
The table below reports adverse reactions by MedDRA system organ classes(MedDRA SOCs).

	Frequency of adverse reactions		
	Common	**Uncommon**	**Rare**
Immune system Disorders			Asthma
Endocrine disorders	Menstrual disorders Intermenstrual bleeding Breast pain		Breast secretion
Nervous system disorders	Headache Depressive mood	Changes in libido	
Ear and labyrinth disorders			Hypacusis
Vascular disease	Migraine	Hypertension Hypotension	Thromboembolism
Gastrointestinal system	Nausea	Vomiting	
Skin and subcutaneous tissue disorders		Acne Eczema Pruritus	
Reproductive system and breast disorder	Leucorrhoea Vaginal moniliasis	Vaginitis	
General disorder and administration site condition		Fluid retention Body weight changes	

The following serious adverse events have been reported in women using COCs, which are discussed in **Special warnings and precautions for use.**

- Venous thromboembolic disorders
- Arterial thromboembolic disorders
- Hypertension.
- Liver tumours.
- Occurrence or deterioration of conditions for which association with COC use is not conclusive: Crohn's disease, ulcerative colitis, epilepsy, migraine, endometriosis, uterine myoma, porphyria, systemic lupus erythematosus, herpes gestationis, Sydenham's chorea, haemolytic uremic syndrome, cholestatic jaundice.
- Chloasma.
- Acute or chronic disturbances of liver function may necessitate the discontinuation of COC use until markers of liver function return to normal.

The frequency of diagnosis of breast cancer is very slightly increased among OC users. As breast cancer is rare in women under 40 years of age the excess number is small in relation to the overall risk of breast cancer. Causation with COC use is unknown. For further information, see **Contraindication and Special warnings and precautions for use**.

OVERDOSE

There has not yet been any experience of overdose with LIZA. On the basis of general experience with combined oral contraceptives, symptoms that may possibly occur in this case are: nausea, vomiting and, in young girls, slight vaginal bleeding. There are no antidotes and further treatment should be symptomatic.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

The contraceptive effect of COCs is based on the interaction of various factors, the most important of which are seen as the inhibition of ovulation and the changes in the cervical secretion.

Pharmacokinetic properties

- *Drospirenone*

Absorption
Orally administered drospirenone is rapidly and almost completely absorbed. Maximum concentrations of the active substance in serum of about 38 ng/ml are reached at about 1 - 2h after single ingestion. Bioavailability is between 76% and 85%. Concomitant ingestion of food has no influence on the bioavailability of drospirenone.

Distribution
After oral administration, serum drospirenone levels decrease with a terminal half-life of 31 h. Drospirenone is bound to serum albumin and does not bind to sex hormone binding globulin (SHBG) or corticoid binding globulin (CBG). Only 3 - 5 % of the total serum concentrations of the active substance are present as free steroid. The ethinylestradiol-induced increase in SHBG does not influence the serum protein binding of drospirenone. The mean apparent volume of distribution of drospirenone is 3.7 ± 1.2 l/kg.

Metabolism
Drospirenone is extensively metabolized after oral administration.The major metabolites in the plasma are the acid form of drospirenone, generated by opening of the lactone ring, and the 4,5-dihydro- drospirenone-3-sulfate, both of which are formed without involvement of the P450 system. Drospirenone is metabolized to a minor extent by cytochrome P450 3A4 and has demonstrated a capacity to inhibit this enzyme and cytochrome P450 1A1, cytochrome P450 2C9 and cytochrome P450 2C19 *in vitro.*

Elimination
The metabolic clearance rate of drospirenone in serum is 1.5 ± 0.2 ml/min/kg. Drospirenone is excreted only in trace amounts in unchanged form. The metabolites of drospirenone are excreted with the feces and urine at an excretion ratio of about 1.2 to 1.4 The half-life of metabolite excretion with the urine and feces is about 40h.

Steady-state conditions
During a treatment cycle, maximum steady-state concentrations of drospirenone in serum of about 70 ng/ml are reached after about 8 days of treatment. Serum drospirenone levels accumulated by a factor of about 3 as a consequence of the ratio of terminal half-life and dosing interval.

Special patient populations
Effect of renal impairment
Steady-state serum drospirenone levels in women with mild renal impairment (creatinine clearance CLcr, 50-80 mL/min) were comparable to those of women with normal renal function. The serum drospirenone levels were on average 37 % higher in women with moderate renal impairment (CLcr, 30 - 50 mL/min) compared to those in women with normal renal function. Drospirenone treatment was also well tolerated by women with mild and moderate renal impairment. Drospirenone treatment did not show any clinically significant effect on serum potassium concentration.

Effect of hepatic impairment
In a single dose study, oral clearance (CL/F) was decreased approximately 50 % in volunteers with moderate hepatic impairment as compared to those with normal liver function. The observed decline in drospirenone clearance in volunteers with moderate hepatic impairment did not translate into any apparent difference in terms of serum potassium concentrations. Even in the presence of diabetes and concomitant treatment with spironolactone (two factors that can predispose a patient to hyperkalemia) an increase in serum potassium concentrations above the upper limit of the normal range was not observed. It can be concluded that drospirenone is well tolerated in patients with mild or moderate hepatic impairment (Child- Pugh B).

Ethnic groups
No clinically relevant differences in the pharmacokinetics of drospirenone or ethinylestradiol between Japanese and Caucasian women have been observed.

- *Ethinylestradiol*

Absorption
Orally administered ethinylestradiol is absorbed rapidly and completely. After administration of 30 µg, peak plasma concentrations of 100 pg/mL are reached 1-2 hours after ingestion. Ethinylestradiol undergoes an extensive first-pass effect, which displays great inter-individual variation. The absolute bioavailability is approx 45 %.

Distribution
Ethinylestradiol has an apparent volume of distribution of 5 L/kg and binding to plasma proteins is approx 98 %. Ethinylestradiol induces the hepatic synthesis of SHBG. During treatment with 30 µg ethinylestradiol the plasma concentration of SHBG increases from 70 to about 350 nmol/L. Ethinylestradiol passes in small amounts into breast milk (0.02 % of the dose).

Metabolism
Ethinylestradiol is metabolized completely (metabolic plasma clearance 5 ml/min/kg).

Elimination
Ethinylestradiol is not excreted in unchanged form to any significant extent. The metabolites of ethinylestradiol are excreted at a urinary to biliary ratio of 4:6. The half-life of metabolite excretion is about 1 day.

Steady-state conditions
Steady-state conditions are reached during the second half of a treatment cycle and serum levels of ethinylestradiol accumulate by a factor of about 1.4 to 2.1

SHELF-LIFE

3 years.

STORAGE

Store below 30°C. Keep out of reach of children. Jauhi daripada kanak-kanak.

PRESENTATION

LIZA is present in a blister of PVC-PVDC/Aluminium which consists of 21 tablets.

MANUFACTURER

LABORATORIOS LEÓN FARMA, S.A.
La Vallina, s/n Polígono Industrial, Navatejera,
24193 León, Spain.

PRODUCT REGISTRATION HOLDER

EURODRUG LABORATORIES (M) SDN BHD
T2-21-10, THE SQUARE, ONE CITY,
Jalan USJ 25/1C, 47650 Subang Jaya,
Selangor, Malaysia.

Date of revision: 18 September 2025